Instructions | Kepler RLHF

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***Use this AFTER reading this entire document* Kepler RLHF | Do's and Don'ts**

This document is meant to be used when you first onboard the project to learn everything you need. You should read this document thoroughly.

🏓 Table of Contents

1️⃣ Section 1: Welcome to Kepler RLHF!

In this project, your work will help improve some of the world's leading AI models! Here, we'll go through the steps of your tasks and explain what you'll be doing in each one.

🍎 **What should you know after finishing this section?**

1. Learn what this project is about
2. Understand what you will have to do in a task

Step 1: Review Your Assigned Language and Locale 📦

- **Language:** You'll be given a specific language and locale.
- **Purpose:** We are teaching the model to answer math questions in a variety of languages so that it can perform well all over the world!
- **Why This Matters**: Covering many topics makes the AI useful for people with different needs and interests. Your careful work here helps make the model more versatile!

Step 2: Write a Prompt ✍️

- **What is a Prompt?:** A prompt is a math question or reasoning problem that you write. It tells the AI what to talk about and how to respond.
- **Key Point:** Your prompt is the foundation of this task.
- **Complexity:** prompts should be high-school/junior college level in complexity
- **Be creative!** Don't always send the same prompt or same topic, be diverse!

Step 3: Review Two AI Responses and Complete Two Jobs 🔍

Now, the AI will respond to your prompt twice. You'll review both responses carefully. You have two main tasks:

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1. **Math-Checking:**
    - **Did the model provide the correct final answer?** Test whether the response provided the correct final answer.
    - **Did the model write out any incorrect steps or make incorrect reasoning steps?** Once the model generates a response to your prompt, review it carefully. Identify areas where the response falls short or makes mistakes. If both of the models are error-free, edit your prompt and try again until you make the model fail
2. **Proofreading:**
    - Think of yourself as the **"language expert."**
    - Check for any unnatural language, awkward phrasing, or grammar errors in your language.
    - The AI might not speak your language perfectly, so look out for things that sound odd or don't flow well.
    - **Did it follow instructions correctly?** Check that the AI stayed within the constraints and didn't make mistakes.
    - **Is the information accurate?** The AI sometimes gets facts wrong, so make sure everything is true and correct.

Step 4: Rate the Responses 🧑‍⚖️

After proofreading and fact-checking, you will rate each response based on how well it performed. Here's what to keep in mind:

- **Provide a Justification:** Your explanation helps the AI understand what it did wrong.
- **Side-by-Side Rating:** If asked, compare both responses and select the better one. Explain why the chosen response is stronger or more accurate. A high-quality explanation helps improve the model.

Step 5: Improve the Selected Response 🎨

Finally, if the selected response has errors, you will have to edit to make it better. Here's what to do:

- **Fix math errors:** correct the incorrect step(s) clearly and concisely. The rewrite should be self-contained and understandable, following a logical sequence of reasoning.
- **Correct Language Errors:** Fix any grammar mistakes, spelling errors, or any fluency

issues.

- **Follow Instructions Precisely:** Ensure that the AI's response matches all instructions perfectly.
- **All Claims are Accurate:** Make sure everything the model says is true and accurate.

Following these steps carefully will make a real difference in improving AI quality. Your attention to detail, accuracy, and fluency help create a model that is more helpful for everyone!

This instructions document will explain every step above in detail to make sure your tasks are high quality!

2️⃣ Section 2: Ground Rules!

To stay on this project, please follow these important rules. Not following them may lead to removal:

- **Math Prompts Only:**
  - This is a math project. The purpose is to ask math or reasoning questions.

- **Reasoning errors not calculation errors:**
  - For this project, each response must include at least one reasoning error. If the only error is a calculation mistake, modify the prompt to create a reasoning error.
  - **Example:** An error where the model makes a mistake in applying a derivative is good. However, asking the model to divide two really large numbers and it messes up a decimal point **is not a good prompt.**

- **Assigned Language Only:**
  - Only use the assigned language for your prompts. No English is allowed in any prompt or response unless specifically required.

- **No External Language Models (LLMs):**
  - Do not use other AI tools, like ChatGPT or Gemini, to generate text or get ideas. These tools often make mistakes, which can lower your work quality.

- **Check for Local Language Quality:**
  - Make sure all responses sound natural and fluent for a native speaker in your language.

- **Carefully Proofread Every Response:**
    - **Fact-check** everything in the responses and make sure the writing quality sounds like a local person who is native in the language you speak in.
    - **Proofread** all spelling, grammar, phrases, flow, ideas, and instruction following components of the response every single time

By following these ground rules, you'll ensure high-quality work and avoid any quality flags. Thank you for helping improve the AI model!

3️⃣ Section 3: Writing Prompts!

🍎 **What should you know after finishing this section?**

1. What is a user prompt?
2. What a good user prompt looks like
3. When to re-try your user prompt?

🧠 **Concept: What is a user prompt?**

A user prompt is a math question or reasoning problem that guides the AI on what to say and how to say it. A good prompt has three elements:

1. **Clear about what the response is asking for:**
    - What it is: The background or main topic that tells the model what the response should answer.
    - The prompt must be clear about what needs to be solved. You must ask the model something that another human would also be able to understand.

2. **Solvable:**
    - Avoid problems that don't contain the information necessary to solve them.
    - Avoid problems with impossible scenarios, for example:
        i. Dividing by zero
        ii. Asking for the square root of a negative number if the domain is real numbers
    - Avoid problems containing terms, concepts, theorems, or lemmas that do not adhere to mathematical rules

3. **One correct solution:**
    - Avoid problems that have many potential answers. We want problems where there is one, well-defined, singular answer.

4️⃣ Section 4: Proofreading and Fact-Checking

🍎 **What should you know after finishing this section?**

1. Understand that you need to be a proofreader
    a. You are an expert on how your language should be spoken and written
    b. The model makes a lot of fluency mistakes, spelling/grammar errors, or awkward phrasing. You need to find all of these!
2. Recognize that you also need to be a math-expert
    a. The model makes many errors in its reasoning, understanding prompt instructions, or solving math questions
    b. Your job is to catch all these errors and fix them **if they're present in the final response**

Here’s how to do this:

Dimensions	Step-by-Step
Instruction Following	**Step 1**: **Review the instructions provided**. Carefully read the original prompt to understand exactly what was asked. Make sure there is **no ambiguity.** **Step 2**: **Compare the response to the instructions and constraints**. Does the response address every part of the prompt? Identify if any parts were ignored or misunderstood. **Step 3**: **Check for adherence to constraints**. If there were specific guidelines (word count, format, structure), verify whether these were followed exactly. **Step 4**: **Determine clarity of adherence**. Assess if the response followed the instructions without misinterpretation or unnecessary additions that deviate from the task.
Truthfulness	**Step 1**: **Identify all math steps and claims**. Highlight any statements of mathematical relevantly claims.. **Step 2**: **Verify the math and reasoning**. Check the accuracy of the math step by cross-referencing reliable sources and using your own knowledge. Use WolframAlpha, a calculator, or Google to check if every claim the model makes is true. The model **almost always makes math**

	or reasoning errors! **Step 3**: **Evaluate correct application**. Ensure that the theorems, properties, or claims facts are applied properly in the context of the response. Misapplied facts can be as misleading as incorrect ones. **Step 4**: **Identify potential errors**. If any part of the response includes inaccurate or misleading information, mark those areas as inaccuracies. **Step 5: Check EVERYTHING.** Even if you find one issue, that doesn't mean the model didn't make another one later on in the response. Don't let inaccurate statements slip by!
Fluency and Localization	**Step 1: Identify the required style/tone.** Determine if the original instruction asked for a specific writing style (e.g., formal, casual) or tone (e.g., friendly, professional). **Step 2: Compare the style/tone.** Check if the writing style and tone match the requirement. If no specific tone was requested, ensure the tone is fluent and **speaks like a native person would**. **Step 3: Evaluate awkward phrasing.** Make sure the tone remains consistent throughout the response. Sudden shifts in tone (e.g., from casual to formal) should be avoided unless explicitly required. The model is usually awkward and might use awkward phrasing or say things that a native person might not necessarily say. **Step 4: Assess clarity and flow.** Ensure that the writing is easy to follow, with clear transitions between ideas and no awkward phrasing. The sentences should flow smoothly and be grammatically correct. The model might use unusual words or make grammar mistakes that don't follow the tenses that are used in your language.

5️⃣ Section 5: Rating Responses

🍎 **What should you know after finishing this section?**

1. What is the role of rating?
2. What do you need to watch out for when rating?

3. What is a justification

🧠 **Concept: Rating**

The rating aspect is simply marking which items the response “checks off”. This is the simplest part of the task, but one that needs the closest attention to detail.

Your rating will be linked to the proofreading and fact-checking we discussed in the previous section. You will do two types of rating:

1. **Dimension Ratings**
    a. Each response will be individually rated on:
        i. Fluency & Localization
        ii. Instruction Following
        iii. Truthfulness
        iv. Writing Style and Clarity
        v. Verbosity
2. **Side-by-side Rating**
    a. You will select the better response and justify why

Any task that doesn’t perform correct ratings risks a low quality score, so please do not rush through this section!

🧠 **Concept: Instruction Following vs Truthfulness**

Important reminder: There may be cases where a model does a great job following instructions, but the response is not truthful, or vice versa.

Think of instruction following and truthfulness as independent dimensions. For example:

- **Truthfulness issue**
  - A response can fully follow the prompts request but give you false content.
- **Instruction following issue**
  - A response can fail to follow the prompts instructions but provide true and factual content.

When assessing the two dimensions, ask yourself:

- Instructions following: Does the response do everything the prompt asked it to do?
- Truthfulness: Is the provided content factual and true?

Common Pitfalls and How to avoid

- ❌Confusing Instruction Following with Truthfulness
  - Assuming that a response that follows instructions well is also factually correct
  - **How to avoid:** Separate your evaluation of how well the response adheres to the prompt from whether the information is accurate.
- ❌Assuming Truthfulness Implies Instruction Following
  - Believing that a factually accurate response must have followed the prompt correctly.
  - **How to avoid:** Check if the response answers the specific question or task in the prompt, regardless of how accurate the information is.
- ❌Assuming Factual Information Is Always Relevant
  - Giving a high IF score because the response contains true statements, even if those statements aren't what the prompt asked for.
  - **How to avoid:** Ensure that the truthfulness of the response is relevant to the prompt's specific question.

🔎 Dimension Ratings

Once you've identified that **one of the responses has errors**, you will need to **rate both responses** on a scale of **1-3** for each criterion, with **3 being the highest score***. Here are the categories to focus on:

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1. **Instruction Following**
    - **Implicit Instructions**: Does the response address implied instructions from the prompt?
    - **Explicit Instructions**: Does the response follow directly stated requirements in the prompt?
    - **Overall Adherence**: Does the response meet all the requirements both implicit and explicit?

2. **Localization**
    - **Cultural Nuances**: Does the response reflect cultural norms and sensitivities of the prompt's context?
    - **Language Adaptation**: Is the response adapted appropriately to the language used in the prompt?

3. **Truthfulness**
    - **Claim Accuracy**: Are all claims and details in the response factually correct?
    - **Logical Steps**: Are reasoning steps valid and error-free?

4. **Response Length**
    - **Optimal Length**: Is the response neither too terse nor overly verbose?
    - **Efficiency**: Does the solution optimize the number of steps while being complete?

5. **Style and Clarity**
    - **Readability**: Is the response easy to read and understand?
    - **Structure**: Is the response well-structured and visually organized?
    - **Formatting**: Does the response use appropriate formatting tools like lists or markdown to enhance clarity?

***Response Length's range is -2 to 2, with 0 being the best part**

⚖️ Side by Side Rating

Selecting the Better Response

After reading both responses, you'll choose which one is better using the following options:

1. **Response A is much better:** Choose this if Response A is significantly better, or if Response B has major issues (e.g., wrong language, gibberish, incomplete).
2. **Response A is better**: Choose this if Response A performs better in most rating criteria, especially the key ones like accuracy, completeness, and fluency.
3. **Response A is slightly better:** Choose this if Response A is only a little better in one or two criteria.
4. **Tie:** Choose this if both responses are of similar quality across all dimensions or have individual strengths that balance out.

The same options apply for Response B if it's better.

Hierarchy of Rating Dimensions

When deciding which response is better, some criteria are more important than others. Here's the priority order:

1. Truthfulness 🎖️ (most important)
2. Instruction Following 🎖️ (also most important)
3. Language Fluency
4. Writing Style & Clarity
5. Verbosity

Writing a Justification for Your Choice

Your justification explains why you chose one response over the other. Here are the key points to keep in mind:

1. Stick to the Evidence: Focus on the main differences between the two responses. No need to mention criteria that don't have issues.
2. Focus on Key Criteria: Only discuss the dimensions that affected your choice (e.g., if Truthfulness was a big difference, mention that).
3. Be Concise: Avoid flowery language or extra details that aren't needed.
4. Depth and Completeness Matter: It's better to focus on the quality and accuracy of information over writing style or formatting.

5. Don’t Use LLMs: Write your justification independently.

6️⃣ Section 6: Fixing + Improving Responses

🍎 **What should you know after finishing this section?**

1. What to fix in a rewrite
2. The importance of having no mathematical or fluency issues whatsoever in the fixed response

After you complete selecting the preferred response and give a justification for your preference score, you will be asked to answer a few questions. These questions will determine if a writing step is needed.

Unless the response you selected is perfect and has no issues at all, **you are expected to fix and improve the selected response so that it doesn’t have any math or language issues**. A perfect response should be accurate and fully aligned with the prompt’s requests and constraints. Fixing a response is composed of three major steps.

Step 1: Truthfulness 🎯

1. **Review claims, math steps, and reasoning**: The response should be mathematically correct and reasoning based on information in the prompt or appropriate mathematical properties/theorem. If there are any errors in the original response, they should be corrected in your perfect response.
2. **How to Rewrite**:
    a. If the original response had incorrect information, rewrite it with the correct information embedded naturally.
    b. **Maintain Sequential Integrity**: Ensure that your rewrite preserves the logical order of the original reasoning and includes the same level of detail.

    c. **Clarity in Rewriting:** Rewrite the step clearly and concisely, using simple language that avoids jargon. The rewrite should be self-contained and understandable, following a logical sequence of reasoning.

d. **Focus on Problem-Solving:** Only include information necessary to solve the problem. Do not add extraneous details like definitions of basic concepts.

3. **What to Look For**: Check that all the facts presented are correct.

Step 2: Instruction Following 📏

- **Compare to the Prompt**: The perfect response should exactly follow the instructions provided in the prompt. This includes adhering to any constraints like format, length, structure, or content requirements.
- **How to Rewrite**: If the initial response deviates from the instructions (e.g., exceeds word count or fails to meet the format), revise it to strictly follow the original guidelines.

Step 3: Improve the response's fluency and style 💬

- Think of yourself as the "language expert."
- Check for any unnatural language, awkward phrasing, or grammar errors in your language.
- The AI might not speak your language perfectly, so look out for things that sound odd or don't flow well.
- Ensure that the model use the tone specified in the prompt and the the tone is respectful of cultural context
- The response should not have **any awkward phrasing or language errors EVER!**

7️⃣ Section 7: Additional Info

How to use the LaTeX generation tool

If you're not familiar with LaTeX and/or would like some assistance in generating LaTeX from your plaintext formulas, we have provided a LaTeX generation tool that can help.

To use this app, first paste your plaintext formulas into the Input Text box and click the "Get LaTeX" button. After a moment, the LaTeX version of your formula will appear below with a

“Copy Text” button that will copy it to your clipboard.

Once you’ve copied the text, return to the outlier text box and paste it in. Make sure to turn on LaTeX rendering so you can see the rendered LaTeX!